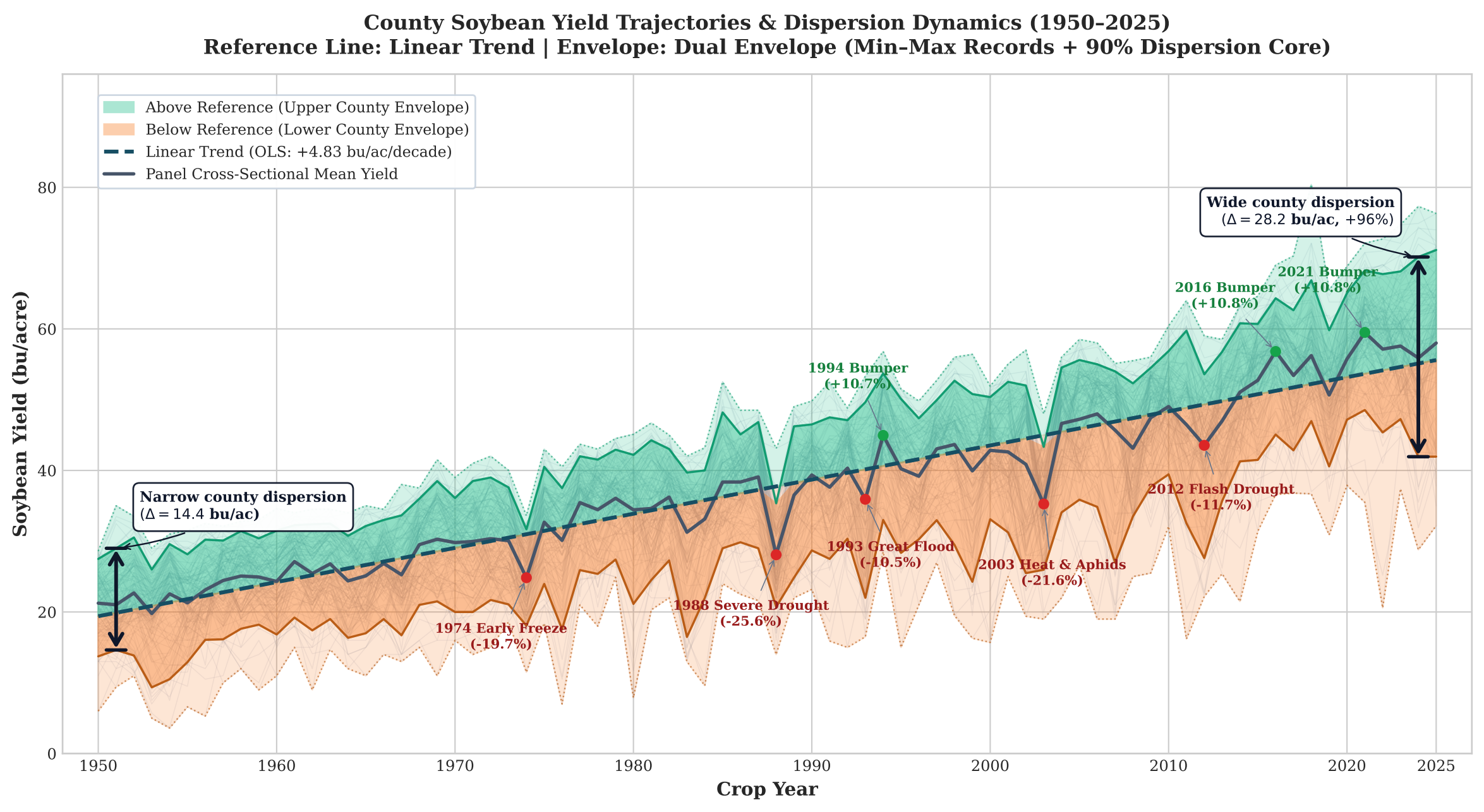

County Soybean Yield Trajectories & Dispersion Dynamics (1950–2025)
Reference Line: Linear Trend | Envelope: Dual Envelope (Min–Max Records + 90% Dispersion Core)
Above Reference (Upper County Envelope)
Below Reference (Lower County Envelope)
Linear Trend (OLS: +4.83 bu/ac/decade)
Panel Cross-Sectional Mean Yield
Soybean Yield (bu/acre)
Crop Year
0
20
40
60
80
1950
1960
1970
1980
1990
2000
2010
2020
2025
Narrow county dispersion
(Δ = 14.4 bu/ac)
Wide county dispersion
(Δ = 28.2 bu/ac, +96%)
1974 Early Freeze
(-19.7%)
1988 Severe Drought
(-25.6%)
1993 Great Flood
(-10.5%)
1994 Bumper
(+10.7%)
2003 Heat & Aphids
(-21.6%)
2012 Flash Drought
(-11.7%)
2016 Bumper
(+10.8%)
2021 Bumper
(+10.8%)